

Project Design Phase
Proposed Solution

Date	10 September 2026
Team ID/SID	SWUID2026136950
Project Name	Automated Network Request Management System (ServiceNow)
Maximum Marks	2 Marks

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Network access requests in most organizations are still handled manually — through emails, chat messages, and phone calls. An employee needing access or a configuration change emails IT or raises an unstructured ticket, which IT then reviews, clarifies, routes for manager approval, and finally assigns to a network engineer. This manual process creates six recurring problems: (1) no standard intake format, so requests arrive inconsistently across email, chat, phone, and tickets, causing missing information and repeated back-and-forth; (2) approvals happen over email or phone with no audit trail and no enforcement before fulfillment; (3) tasks are created and assigned manually, often with incomplete context, forcing engineers to follow up; (4) requesters have no visibility into status and must chase updates themselves; (5) managers cannot measure turnaround time, team load, or SLA compliance; and (6) audit preparation takes weeks because there is no centralized log of approvals and actions. The business impact is measurable: average fulfillment time of 3–5 days, 30–45 minutes of manual effort per request, approval delays affecting roughly 40% of requests, and frequent audit findings in regulated environments. The core problem is the absence of a standardized, automated, and traceable system for managing network requests end to end — which this project solves by building a complete workflow on ServiceNow.

2	Idea / Solution description	Overview. The solution is an automated Network Request Management application built on ServiceNow, centered on a Service Catalog item where employees submit requests with structured details — Request Type, Device, Access Level, and Business Justification — instead of free-form emails. Smart, adaptive form. Two dynamic UI Policies tailor the form to the selected Request Type: choosing Firewall shows and mandates Device and Port; choosing Access shows only Access Level; choosing Configuration shows Device and Port but leaves Port optional. This keeps the form short and prevents incomplete submissions. Data model. A custom table stores each request's type, access level, device, port, justification, state, and requesting user. Business rules automatically default the state to New and set "requested for" to the current user, reducing manual data entry.
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		Automated workflow. Submission triggers a Flow Designer workflow that runs without manual intervention: it routes the request to the Network Approvers group and waits for a decision; on approval, it creates a Catalog Task assigned to the Network Team; it waits for the engineer to mark the task Closed Complete; and it then automatically closes the Requested Item. Notifications. Four email notifications keep every stakeholder informed — submission confirmation to the requester, an approval request with a direct action link to the approver, an approved/rejected notice to the requester, and a completion notice once the request is fulfilled. Governance. Access is controlled through three roles (requester, approver, engineer) and three groups (Network Requesters, Network Approvers, Network Team), with ACLs enforcing read, create, and write permissions at the record level. Together, these pieces deliver a fully automated, end-to-end workflow — from submission to closure — with a complete audit trail and real-time visibility for every stakeholder.

3	Novelty / Uniqueness	• Dynamic form behavior — fields appear only when relevant to the selected request type, improving data quality and cutting incomplete submissions. • True end-to-end automation — not just a form or a workflow, but a single continuous process from submission through approval, task creation, fulfillment, and closure. • Approval-based routing — the flow pauses for approval before proceeding, enforcing governance automatically with a full "who approved what, when" audit trail. • Auto-task assignment — an approved request creates and assigns a Catalog Task to the Network Team without any manual coordination. • Closed-loop automation — engineer task closure automatically closes the Requested Item and notifies the requester, closing the loop with zero manual steps. • Real-time, role-aware notifications — four lifecycle emails, including a direct action link for approvers, keep every stakeholder current. • Role-based security — three roles and three groups, enforced through ACLs at the record level, restrict who can submit, approve, or fulfill. • Custom scoped application — the solution is self-contained, modular, and deployable independently of other platform customizations. • Audit-ready by design — every action, approval, and state change is logged automatically, cutting audit preparation time drastically. • Built-in reporting — dashboards give real-time visibility into request volume, task performance, and SLA compliance.
4	Social Impact / Customer Satisfaction	Social impact. The system frees IT teams from manual coordination so they can focus on real technical work, reduces stress and follow-up fatigue, and provides the audit trail that regulated industries — finance, healthcare, government — require. It also cuts errors from unclear requests and makes IT more transparent and accountable to the rest of the organization.
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		Customer satisfaction. Requesters get faster fulfillment and real-time status visibility without needing to chase updates. Approvers get clear, one-click actionable notifications. Network engineers receive complete context up front and can start work immediately. IT managers gain visibility into volume, turnaround, and SLA compliance for better decision-making, and auditors get a reliable, complete trail for every change. Overall, the solution raises the experience for every participant in the network request lifecycle — not just the requester.

5	Business Model (Revenue Model)	The solution runs on ServiceNow, which the organization already licenses, so it carries no additional platform cost. • Cost savings — roughly 30 minutes of manual coordination saved per request; for an organization processing 100 requests/week, that is about 50 hours of effort saved weekly. • Time savings — average fulfillment time drops from 3–5 days to under 1 day, improving organization-wide productivity. • Compliance savings — automatic, centralized logging of approvals and changes cuts audit preparation time by up to 80%. • Scalability — handling more requests adds no incremental cost, since the system runs on existing infrastructure. For ServiceNow partners, the application can be offered to clients as a packaged, scoped application on the ServiceNow Store, extended with premium SLA analytics, dashboards, and change-management integration, and adapted to other IT request types such as server, database, or application access.
6	Scalability of the Solution	Built on ServiceNow's cloud platform, the solution is designed for enterprise-scale use from day one. • Volume — supports 100+ requests per day with reliable handling of concurrent submissions and flow executions. • Data scale — the custom table holds thousands of records without performance degradation. • Concurrency — Flow Designer processes multiple requests in parallel without conflicts. • Visibility at scale — reports and dashboards stay real-time as request volume grows. • Availability — multi-region ServiceNow Cloud hosting keeps the system accessible even during regional outages, with no extra infrastructure required. Roadmap. Planned extensions include multi-level approvals for high-risk requests, change-management integration for firewall changes, Virtual Agent support for natural-language intake, a mobile-friendly portal, and expansion to other IT request types beyond networking — so the system scales with the organization without rework.